MOH'S P_3 CURVE IS A GLOBAL SET-THEORETIC COMPLETE INTERSECTION

Abstract. A classical question asks whether every irreducible affine curve $C' \subseteq \mathbb{A}^3_{\mathbb{C}}$ can be written set-theoretically as $C' = V(f, g)$ for two polynomials. We do not resolve that problem in general. We prove the global two-equation statement for the singular Moh curve

$$t \longmapsto (t^6 + t^{31}, t^8, t^{10}).$$

Let k be a field of characteristic zero, let

$$\rho : k[x, y, z] \longrightarrow k[t], \qquad (x, y, z) \longmapsto (t^6 + t^{31}, t^8, t^{10}),$$

and put $\mathfrak{p} = \ker \rho$. We construct two explicit degree-16 polynomials $H_1, H_2 \in \mathbb{Z}[x, y, z]$ and prove the containment

$$\mathfrak{p}^2 \subseteq (H_1, H_2) \subseteq \mathfrak{p}.$$

Consequently $\sqrt{(H_1, H_2)} = \mathfrak{p}$ in the global polynomial ring and $\operatorname{ara}(\mathfrak{p}) = 2$. In particular the corresponding formal prime in $k[[x, y, z]]$ is a set-theoretic complete intersection, answering the formal-local question recorded by D'Cruz. The construction starts from the Hilbert–Burch matrix of $\mathfrak{p}$: a polynomial row operation of constant determinant converts its first three rows into a symmetric 3×3 block. A determinant and an adjugate quadratic form then give H_1, H_2, while a symmetric adjugate identity gives the square containment. The ancillary files verify the parametrization kernel and the polynomial identities used in the two containments by exact arithmetic.

1. Introduction

The problem that motivates this paper is the following. Let $C' \subseteq \mathbb{A}^3_{\mathbb{C}}$ be an irreducible affine curve and let $I(C') \subseteq \mathbb{C}[x, y, z]$ be its defining prime ideal. Since $I(C')$ has height two, one may ask whether there always exist two polynomials $f, g \in \mathbb{C}[x, y, z]$ such that

$$C' = V(f, g), \qquad \text{equivalently} \qquad I(C') = \sqrt{(f, g)}.$$

In terms of arithmetic rank, this asks whether every such height-two prime has $\operatorname{ara}(I(C')) = 2$. This global affine-space question is open in characteristic zero for arbitrary singular curves; see the surveys of Lyubeznik and D'Cruz [7, 8, 4].

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We treat the concrete singular curve

$$C = \{(t^6 + t^{31}, t^8, t^{10}) : t \in k\} \subseteq \mathbb{A}^3_k,$$

with defining prime

$$\mathfrak{p} = \ker\bigl(k[x, y, z] \longrightarrow k[t]\bigr), \qquad (x, y, z) \longmapsto (t^6 + t^{31}, t^8, t^{10}).$$

Completing at the origin gives the formal prime

$$P = \ker\bigl(k[[x, y, z]] \longrightarrow k[[t]]\bigr),$$

for the same parametrization. D'Cruz explicitly asked whether this formal prime is a set-theoretic complete intersection [4, Question 2.5]. Moh's family provides height-two primes that can require arbitrarily many ideal-theoretic generators [10]. For the present P_3, Gonzalez and Planas-Vilanova proved that four generators are minimal in characteristic zero [6]. Thus this example is not an ideal-theoretic complete intersection.

Our main theorem gives the following global containment.

Theorem 1.1. *Let k be a field of characteristic zero and let*

$$S = k[x, y, z], \qquad \mathfrak{p} = \ker\bigl(S \longrightarrow k[t]\bigr),$$

where $(x, y, z) \mapsto (t^6 + t^{31}, t^8, t^{10})$. The explicit polynomials H_1, H_2 of Appendix A satisfy

$$\mathfrak{p}^2 \subseteq (H_1, H_2) \subseteq \mathfrak{p}.$$

Consequently

$$\sqrt{(H_1, H_2)} = \mathfrak{p} \qquad \textit{and} \qquad \operatorname{ara}_S(\mathfrak{p}) = 2.$$

The formal-local statement is a corollary.

Corollary 1.2. *Let $R = k[[x, y, z]]$ and let P be the kernel of the same parametrization into $k[[t]]$. Then*

$$\sqrt{(H_1, H_2)R} = P.$$

In particular $\operatorname{ara}_R(P) = 2$.

Proof. It remains to justify that the formal kernel is the extended polynomial kernel. Put

$$B = S/\mathfrak{p} = k[t^6 + t^{31}, t^8, t^{10}] \quad \text{and} \quad \mathfrak{n} = (x, y, z)B.$$

The ring $k[t]$ is finite over B, since t is integral over B by $t^8 = y$. Moreover $\operatorname{Frac}(B) = k(t)$: in $\operatorname{Frac}(B)$ one has

$$t^2 = \frac{z}{y}, \qquad t^6 = \left(\frac{z}{y}\right)^3, \qquad t^{31} = x - \left(\frac{z}{y}\right)^3, \qquad t = \frac{t^{31}}{(t^2)^{15}}.$$

Thus the normalization of $B_\mathfrak{n}$ is $k[t]_{(t)}$; the prime (t) is the unique prime above $\mathfrak{n}$, since $t^8 = y \in \mathfrak{n}$. The finite injection

$$B_\mathfrak{n} \longrightarrow k[t]_{(t)}$$

remains injective after $\mathfrak{n}$-adic completion. On the right, $\mathfrak{n}k[t]_{(t)} = (t^6)$ because $t^6 + t^{31} = t^6(1 + t^{25})$, so this completion is $k[[t]]$. On the left it is $R/\mathfrak{p}R$. The resulting injection is precisely the formal parametrization, and hence $P = \mathfrak{p}R$.

The inclusions in Theorem 1.1 remain true after extension from S to R. Taking radicals now gives the claim. □

The proof of Theorem 1.1 uses a symmetric presentation of the Hilbert–Burch matrix. Section 2 recalls the four generators and their Hilbert–Burch matrix. Section 3 gives a row operation with constant determinant 216 that converts the top 3×3 block into a symmetric matrix M. If r is the last row, then the maximal minors are generated by

$$\Delta = \det M, \qquad v = \operatorname{adj}(M) r^{\mathsf{T}}.$$

The two equations are scalar multiples of

$$\Delta \qquad \text{and} \qquad \Gamma = r \operatorname{adj}(M) r^{\mathsf{T}}.$$

A 3×3 symmetric adjugate identity shows that every product $v_i v_j$ lies in (Δ, Γ), hence the square of the prime lies in the two-generated ideal. No localization, saturation, or primary decomposition is required.

2. The four generators and the Hilbert–Burch matrix

Throughout, $S = k[x, y, z]$ with $\operatorname{char} k = 0$. The characteristic-zero generators from [6] are

$$\begin{aligned}
f_1 &= x^4 - x^2y^4z^3 - 2xy^6z^2 - 4xyz - 3y^3z^5 + 3y^3,\\
f_2 &= x^3y - xy^5z^3 - 3xz^2 - 2y^7z^2 + 2y^2z,\\
f_3 &= 2x^3z - 3x^2y^2 - 2xy^4z^4 - y^6z^3 + yz^2,\\
f_4 &= x^2yz - 2xy^3 - y^5z^4 + z^3.
\end{aligned}$$

Thus

$$\mathfrak{p} = (f_1, f_2, f_3, f_4).$$

Consider the 4×3 matrix

$$\Phi = \begin{pmatrix} 0 & 2xy & -2z(2xy^3z^2 + y^5z - 1) \\ z & -2x^2 & y(4x^2yz^3 + 2xy^3z^2 - 3) \\ -2y & -3z & -x + 6y^2z^4 \\ 3x & 3y & 0 \end{pmatrix}.$$

Lemma 2.1. *The signed* 3×3 *minors of* Φ*, obtained by deleting rows* $1, 2, 3, 4$*, are respectively*

$$6f_1, -6f_2, 6f_3, -6f_4.$$

Consequently $I_3(\Phi) = \mathfrak{p}$*, and* Φ *is a Hilbert–Burch matrix for* $\mathfrak{p}$*.*

Proof. The minor identities are direct polynomial calculations. Since $\mathfrak{p}$ is a height-two prime and $6 \in k^\times$, the ideal of maximal minors is exactly $\mathfrak{p}$. The Hilbert–Burch theorem gives the asserted presentation; see [5, Theorem 20.15]. $\square$

3. A constant-determinant symmetrization

Set

$$\alpha = 16x^3yz^2 - 10x^2y^3z + 18y^2z^3, \qquad \beta = 16x^2y^2z^2 - 10xy^4z,$$

and define

$$L = \begin{pmatrix} 0 & 4 & 0 & 0 \\ -9 & 0 & 0 & -\frac{8}{3}x \\ \alpha & \beta & 6 & 6y^4z^2 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

A direct calculation gives

$$\det L = 216.$$

Hence $L \in \mathrm{GL}_4(S)$. Multiplying the Hilbert–Burch matrix on the left gives

$$L\Phi = \begin{pmatrix} M \\ r \end{pmatrix}, \qquad r = (3x, 3y, 0),$$

where M is the symmetric matrix

$$M = \begin{pmatrix} 4z & -8x^2 & 4y(4x^2yz^3 + 2xy^3z^2 - 3) \\ -8x^2 & -26xy & 18z(2xy^3z^2 + y^5z - 1) \\ 4y(4x^2yz^3 + 2xy^3z^2 - 3) & 18z(2xy^3z^2 + y^5z - 1) & m_{33} \end{pmatrix},$$

with

$$m_{33} = 2\big(16x^3yz^3 - 34x^2y^3z^2 - 36xy^5z^6 + 15xy^5z - 3x - 18y^7z^5 + 36y^2z^4\big).$$

Because left multiplication by L is an automorphism of S^4, the ideal of maximal minors is unchanged:

$$I_3\begin{pmatrix} M \\ r \end{pmatrix} = I_3(\Phi) = \mathfrak{p}.$$

Put

$$\Delta = \det M, \qquad v = \mathrm{adj}(M)r^\mathsf{T} = (v_1, v_2, v_3)^\mathsf{T}, \qquad \Gamma = r\,\mathrm{adj}(M)r^\mathsf{T}.$$

Up to signs, the four maximal minors of the 4×3 matrix $\left(\begin{smallmatrix} M \\ r \end{smallmatrix}\right)$ are v_1, v_2, v_3, Δ. Hence

$$\mathfrak{p} = (\Delta, v_1, v_2, v_3). \tag{1}$$

The following elementary identity is the algebraic core of the proof.

Lemma 3.1 (symmetric adjugate identity)**.** *Let A be any commutative ring, let $M \in M_3(A)$ be symmetric, and let $r = (r_1, r_2, r_3)$. Put*

$$\Delta = \det M, \qquad C = \mathrm{adj}(M), \qquad v = Cr^\mathsf{T}, \qquad \Gamma = rCr^\mathsf{T},$$

and let

$$R_r = \begin{pmatrix} 0 & -r_3 & r_2 \\ r_3 & 0 & -r_1 \\ -r_2 & r_1 & 0 \end{pmatrix}.$$

Then

$$vv^{\mathsf{T}} - \Gamma C = -\Delta\, R_r M R_r^{\mathsf{T}}. \tag{2}$$

Consequently $v_i v_j \in (\Delta, \Gamma)$ *for all* $1 \le i, j \le 3$.

Proof. Equation (2) is the 3×3 form of Jacobi's complementary-minor identity. One may verify it over the universal polynomial ring in the six independent entries of a symmetric M and the three entries of r: for $\Delta \neq 0$, substitute $C = \Delta M^{-1}$ and apply Jacobi's formula for the 2×2 minors of M^{-1}; clearing denominators gives (2), hence the polynomial identity holds universally. Entrywise, the left side therefore differs from a multiple of Γ by a multiple of Δ, which gives the final assertion. $\square$

4. The two global equations

Define

$$H_1 = -\frac{1}{16}\Delta, \qquad H_2 = \frac{1}{36}\Gamma. \tag{3}$$

The determinant calculation shows that these are in fact polynomials with integer coefficients; their expanded forms are recorded in Appendix A. Both have degree 16.

Proposition 4.1. *The polynomials in* (3) *satisfy*

$$\mathfrak{p}^2 \subseteq (H_1, H_2) \subseteq \mathfrak{p}.$$

Proof. By (1), $\Delta \in \mathfrak{p}$ and each $v_i \in \mathfrak{p}$. Since

$$\Gamma = r_1 v_1 + r_2 v_2 + r_3 v_3,$$

we also have $\Gamma \in \mathfrak{p}$. Thus $(H_1, H_2) = (\Delta, \Gamma) \subseteq \mathfrak{p}$, because 16 and 36 are units.

For the reverse containment, (1) shows that $\mathfrak{p}^2$ is generated by products among Δ, v_1, v_2, v_3. Every product containing Δ lies in (Δ, Γ), and Lemma 3.1 gives

$$v_i v_j \in (\Delta, \Gamma) \qquad (1 \le i, j \le 3).$$

Hence $\mathfrak{p}^2 \subseteq (\Delta, \Gamma) = (H_1, H_2)$. $\square$

Proof of Theorem 1.1. Proposition 4.1 gives

$$\mathfrak{p}^2 \subseteq (H_1, H_2) \subseteq \mathfrak{p}.$$

Since $\mathfrak{p}$ is prime,

$$\mathfrak{p} = \sqrt{\mathfrak{p}^2} \subseteq \sqrt{(H_1, H_2)} \subseteq \sqrt{\mathfrak{p}} = \mathfrak{p}.$$

Thus $\sqrt{(H_1, H_2)} = \mathfrak{p}$. Since $\operatorname{ht}\mathfrak{p} = 2$, the arithmetic rank is exactly two. $\square$

Remark 4.2. The containment identities in Section 6 use only integer coefficients and scalar factors divisible by 2 and 3. Combined with the generator theorem of [6], the same global conclusion therefore persists in characteristic $p \geq 5$. We state the main theorem in characteristic zero because that is the case of the classical open question considered here.

5. Comparison with the local construction

The global equations H_1, H_2 are different from the degree-27 and degree-22 pair obtained from the conormal/canonical-module construction in the earlier local argument. That local pair correctly defines the formal germ at the origin but possesses the additional global component

$$(x, 4z^5 - 1).$$

For the present pair, the global containment

$$\mathfrak{p}^2 \subseteq (H_1, H_2) \subseteq \mathfrak{p}$$

precludes additional minimal primes. No localization or saturation is required.

The proof uses a structural feature of the Hilbert–Burch matrix. The constant-determinant row operation L turns three of its rows into a symmetric matrix. For a 4×3 matrix of the form $\left(\begin{smallmatrix} M \\ r \end{smallmatrix}\right)$ with M symmetric, the determinant $\det M$ and the adjugate quadratic form $r \operatorname{adj}(M) r^{\mathsf{T}}$ automatically generate an ideal containing the square of the maximal-minor ideal. In the Moh example the symmetrization occurs globally over the polynomial ring because $\det L = 216$ is constant.

6. Computational verification

The proof does not require a direct radical computation. The explicit polynomial identities can also be checked by exact computer algebra.

Over $\mathbb{Q}$, the ancillary verification package performs the following checks.

(i) It recomputes the kernel of

$$\mathbb{Q}[x, y, z] \longrightarrow \mathbb{Q}[t], \qquad (x, y, z) \longmapsto (t^6 + t^{31}, t^8, t^{10}),$$

by elimination and verifies that it equals (f_1, f_2, f_3, f_4).

(ii) It verifies $\det L = 216$ and the exact matrix identity

$$L\Phi = \begin{pmatrix} M \\ r \end{pmatrix}, \qquad M = M^{\mathsf{T}}.$$

(iii) It verifies

$$\det M = -16 H_1, \qquad r \operatorname{adj}(M) r^{\mathsf{T}} = 36 H_2.$$

(iv) It verifies the short membership identities

$$\begin{aligned} H_1 &= -24x f_1 + \left(64x^4 y z^2 - 40x^3 y^3 z + 72x y^2 z^3 - 81 y^4 z^2\right) f_3 + 81 f_4, \\ H_2 &= -12y f_1 + 27x f_2 + \left(-40x^3 y^2 z^2 + 25x^2 y^4 z + 36 y^3 z^3\right) f_3, \end{aligned}$$

which directly prove $(H_1, H_2) \subseteq \mathfrak{p}$.

(v) For each $1 \le i \le j \le 4$, it supplies integer polynomials A_{ij}, B_{ij} and a positive integer d_{ij} such that
$$d_{ij} f_i f_j = A_{ij} H_1 + B_{ij} H_2.$$
The ten scalar factors are

(i,j)	11	12	13	14	22	23	24	33	34	44
d_{ij}	5832	2916	972	8748	1458	486	4374	162	1458	13122.

As a compact example,
$$162 f_3^2 = -y(5x^3 - 2yz)H_1 - x(8x^3 + 13yz)H_2.$$
These ten identities prove $\mathfrak{p}^2 \subseteq (H_1, H_2)$ over characteristic zero.

The supplied checker uses sparse rational polynomial arithmetic and verifies the displayed identities independently of the script that produced them. A Macaulay2 replay file is also included so that the kernel computation and the ideal containments can be reproduced in a standard commutative-algebra system [9]. No floating-point computation or numerical root finding is used.

Appendix A. Expanded global equations

We record the two global equations in expanded form:
$$\begin{aligned} H_1 = {} & 128x^7yz^3 - 272x^6y^3z^2 - 128x^5y^5z^6 + 120x^5y^5z - 24x^5 \\ & + 16x^4y^7z^5 + 208x^4y^2z^4 + 40x^3y^9z^4 - 394x^3y^4z^3 \\ & - 144x^2y^6z^7 + 291x^2y^6z^2 + 177x^2yz + 90xy^8z^6 \\ & + 144xy^3z^5 - 234xy^3 + 81y^{10}z^5 - 162y^5z^4 + 81z^3, \end{aligned}$$

$$\begin{aligned} H_2 = {} & -80x^6y^2z^3 + 170x^5y^4z^2 + 80x^4y^6z^6 - 75x^4y^6z + 15x^4y \\ & - 10x^3y^8z^5 + 32x^3y^3z^4 - 25x^2y^{10}z^4 - 98x^2y^5z^3 - 81x^2z^2 \\ & - 72xy^7z^7 - 30xy^7z^2 + 102xy^2z - 36y^9z^6 + 72y^4z^5 - 36y^4. \end{aligned}$$

Appendix B. Verification files

The ancillary package contains the expanded equations, the ten square-containment identities, machine-readable coefficient data, a standalone exact-arithmetic checker, and a Macaulay2 replay script. The radical equality follows directly from the two ideal containments and does not require a primary decomposition.

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